

RIYA PALESHA

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Education

University of Massachusetts Amherst

May 2028

MSc Computer Science

Pune Institute of Computer Technology

2020 – 2024

B.E. Computer Engineering

GPA : 9.53

Indian Institute of Technology, Madras

2021 – 2026

BSc Programming and Data Science

GPA : 8.70

Technical Skills

Programming Languages : Python, C++, JavaScript, SQL, HTML/CSS

Backend & Web Technologies : REST APIs, Node.js, FastAPI, Flask, React, Next.js, Vue.js, Express

Cloud & DevOps : AWS, GitLab CI/CD, Jenkins, Grafana, Prometheus

Databases & Storage : PostgreSQL, MongoDB, Aerospike

Core Concepts : Object-Oriented Programming (OOP), Data Structures & Algorithms, Distributed Systems, AI/Machine Learning

Developer Tools : Git, GitHub, Jira, Postman, VS Code, Eclipse

Work Experience

Barclays | Software Engineer - BA4

July 2024 – Aug 2026

- Optimized system observability across distributed microservices, **reducing mean time to detect (MTTD) operational incidents by 45%**, by implementing real-time Grafana monitoring dashboards and Prometheus alert pipelines.
- Architected and deployed highly available, multi-region AWS disaster recovery solutions for the Fraud Platform, ensuring zero downtime during regional outages through modular CloudFormation templates.
- Managed complex quarterly database restacking operations across 51 EC2 nodes, ensuring flawless execution with **zero data loss or service downtime** across environments.
- Automated Aerospike cluster configuration workflows, **cutting deployment errors by over 70%** and eliminating manual intervention, by building GitLab CI/CD pipelines and IaC scripts.
- Streamlined infrastructure provisioning across development and production environments, **reducing setup latency by 80%**, by developing modular CloudFormation Templates, AWS Lambda functions and Gitlab pipelines.
- Spearheaded infrastructure setup and IAM access controls for an AWS MSK (Kafka) cluster, configuring scoped roles and custom alerts that enabled **5+ application teams** to securely deploy operations.
- Eliminated hardcoded database credentials by architecting the aerospike-secret-agent service, automating secret retrieval from a central vault to AWS and injecting credentials securely into database configurations via Chef recipes.

Barclays | Summer Intern

June 2023 – Aug 2023

- Automated enterprise Jira governance workflows, **cutting manual project tracking effort by over 60%**, by scripting Groovy automation tools integrated with Jenkins CI/CD pipelines.
- Designed responsive UI components for the Fraud Platform Dashboard using React, enhancing usability and operational visibility for internal teams.

Projects

ASKIVA: AI-Powered Academic Chatbot | Next.js, Tailwind CSS, Flask, SQLAlchemy, ChromaDB, LangChain

March 2025

- Architected ASKIVA, an AI-powered academic chatbot using LangChain, Google generative models, and ChromaDB to automate doubt resolution, reducing student response times and improving concept comprehension across user cohorts.
- Engineered a secure Flask and SQLAlchemy backend with JWT authentication and role-based Next.js dashboards, enabling students to access context-aware learning workflows while empowering administrators to manage course content and monitor engagement.

Data-Driven Business Performance Analysis | Python, EDA, Data Visualization, Business Analytics

October 2024

- Conducted exploratory data analysis across **2.5+ years of financial and inventory records**, evaluating transactional ledgers for **10 clients** and inventory movement across **7 product categories** to identify high-demand SKUs and cash flow leakages.
- Delivered actionable inventory and payment restructuring strategies that optimized stock turnover ratios and addressed key service delivery pain points identified through customer sentiment feedback.

Legal Document Understanding & Query Assistant | React, FastAPI, MongoDB, PyPDF2, LangChain

January 2024

- Co-authored peer-reviewed research on legal document NLP and developed a full-stack platform (React/FastAPI) that empowers non-legal users to easily query, navigate, and comprehend complex legal documents.
- Developed a semantic Q&A engine with LangChain, PyPDF2 text extraction, custom chunking and a MongoDB-backed document diffing system to detect contract revisions.

Research Work

Reducing Hallucinations through Enhanced Retrieval-Augmented Generation Pipelines

- Engineered a document Q&A evaluation framework across **250+ benchmark queries** using Ollama, LangChain, and confidence scoring, **reducing LLM hallucination rates to <2%**.
- Achieved 94% factual retrieval precision** across multi-format documents (PDF, CSV, DOCX) by implementing semantic chunking, hierarchical partitioning, query reformulation, reranking, and FAISS/ChromaDB vector indexing.